External Supplement to Synthesizing Signaling Pathways from Temporal Phosphoproteomic Data

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Figures

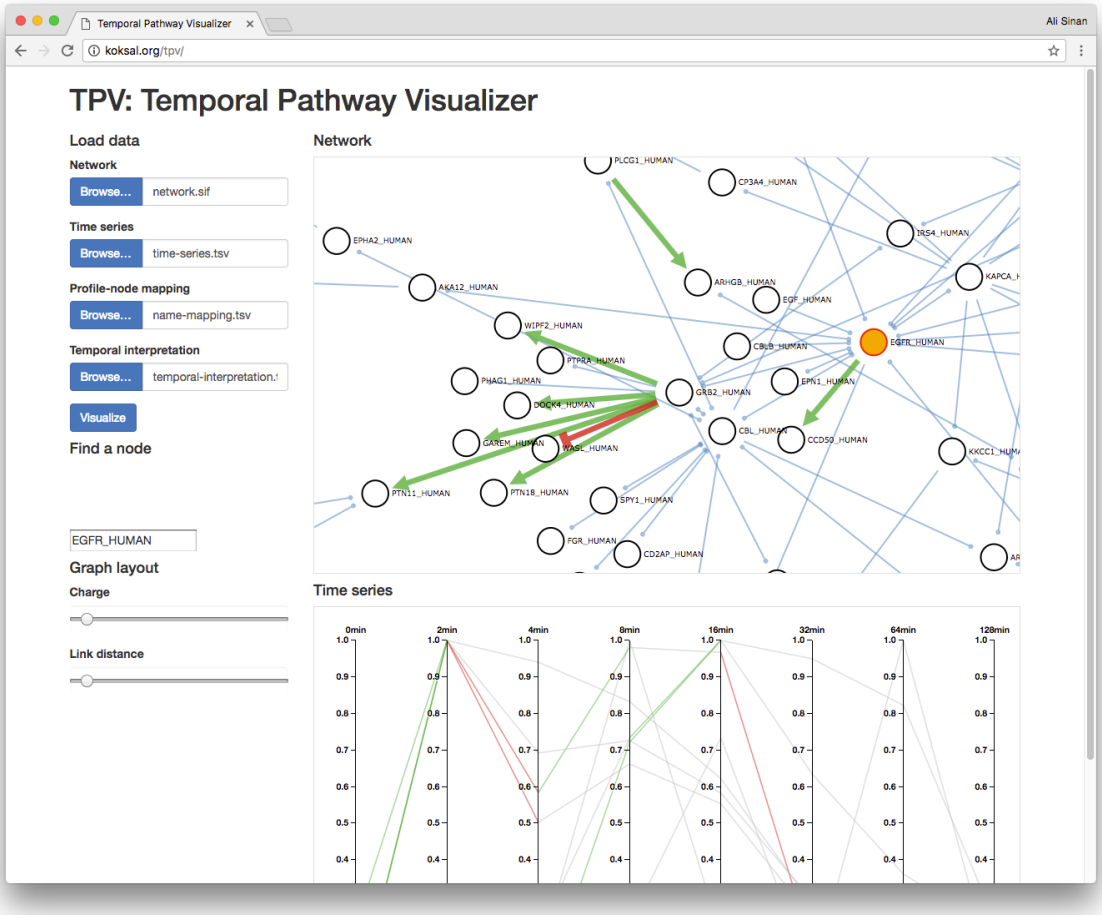


Figure 1. The Temporal Pathway Visualizer (https://github.com/koksal/tpv) supports exploring a network and time series data interactively. Users can select network nodes to filter the time series view or select time series data to highlight the corresponding network nodes.

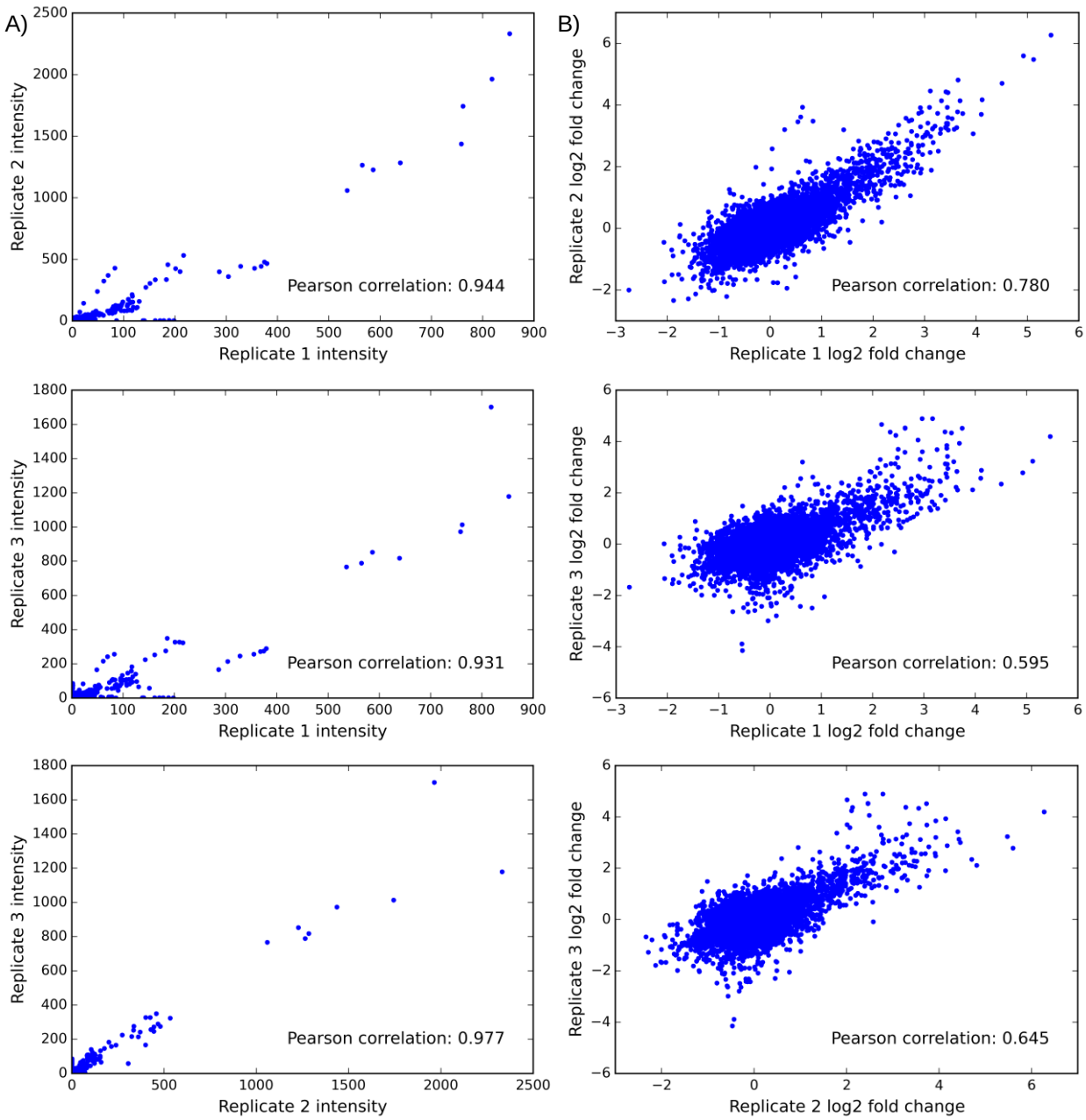


Figure 2. A) The phosphorylation intensities in pairs of biological replicates for all 1,068 phosphopeptides detected in all three replicates. The phosphorylation at each time point is displayed separately. Examining the replicate correlation of $\log_{10}$(intensity + 1) reduces the dominance of high-intensity phosphopeptides and confirms consistency across replicates. The

Pearson's correlation coefficients for the log-scale intensities are 0.783, 0.767, and 0.838. B) The $\log_2$ phosphorylation fold changes for the 1,058 phosphopeptides that did not have any intensities of zero. Fold changes with respect to the 0 minute time point and the previous time point are depicted.

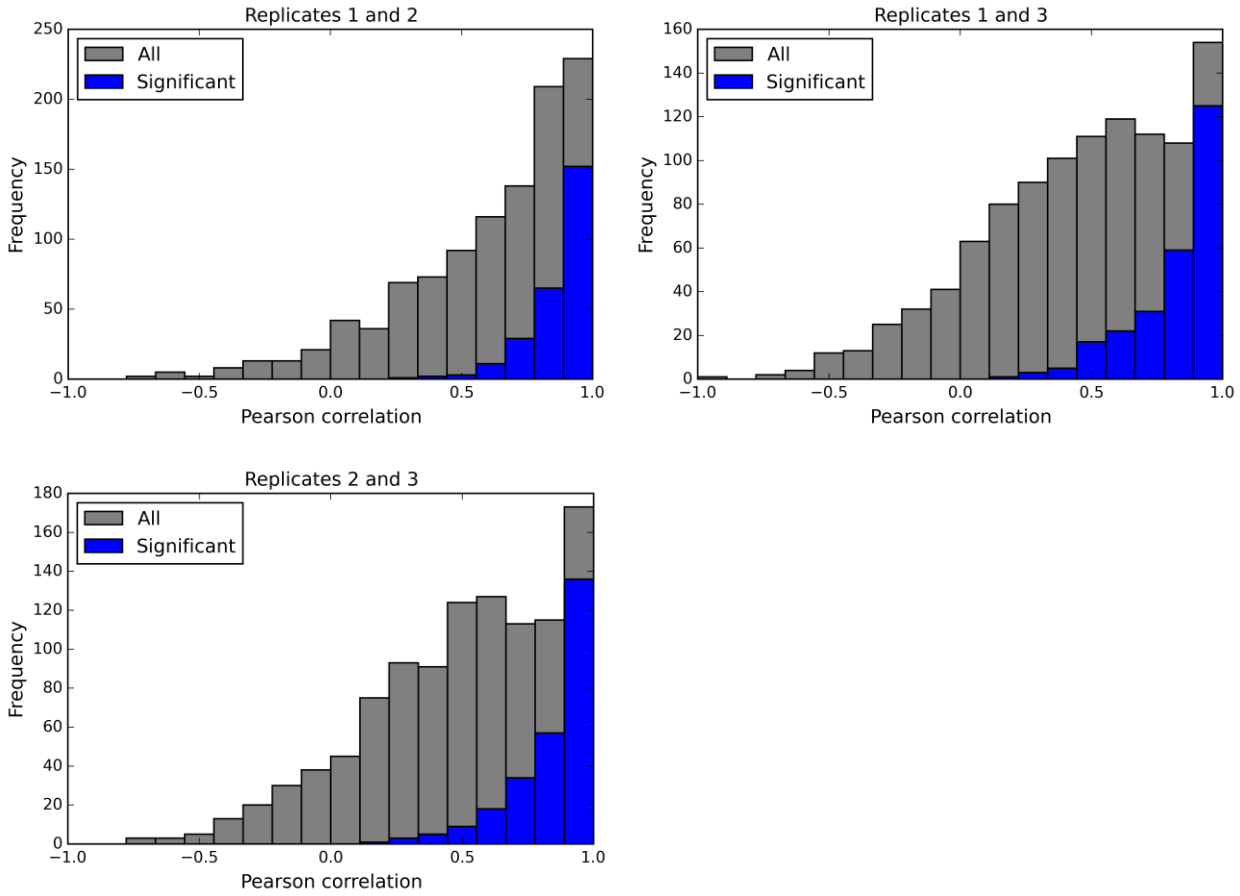
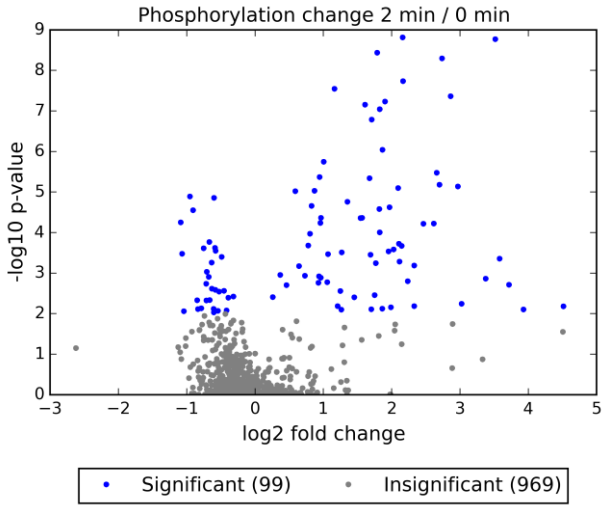
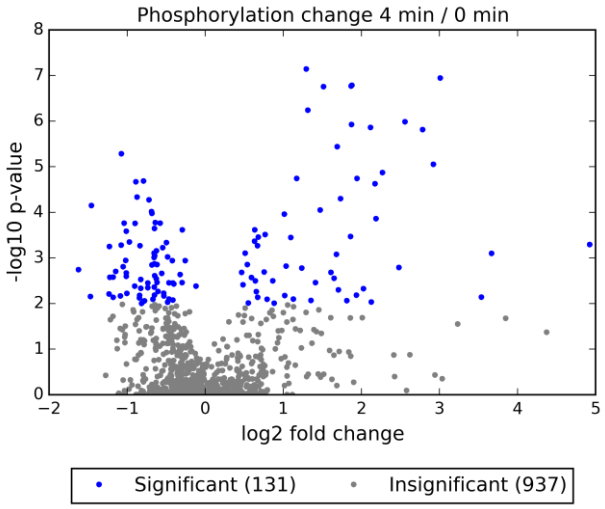
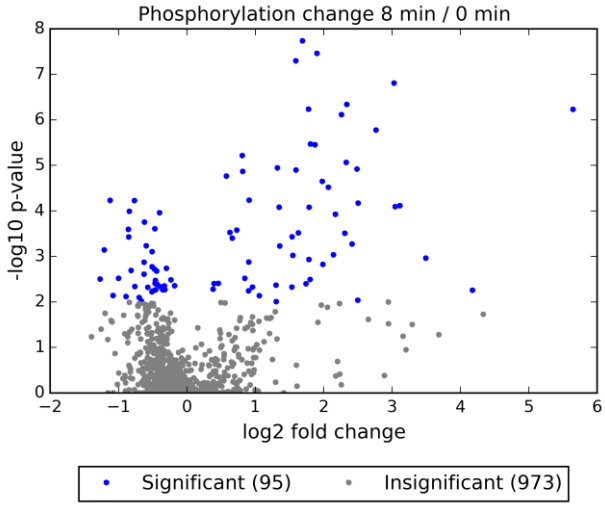
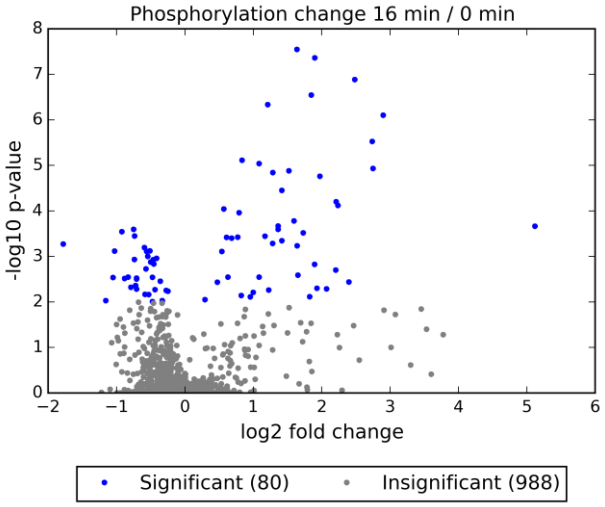


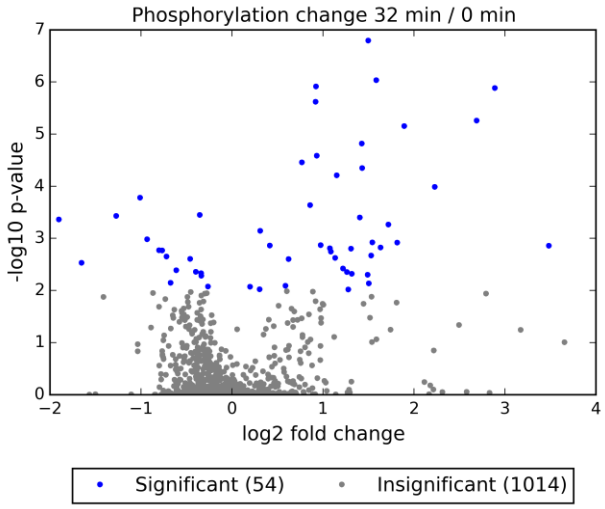
Figure 3. The histograms show the distribution of Pearson's correlation coefficient for the time series profiles in pairs of biological replicates. Phosphorylation intensity correlation for each of the 1,068 phosphopeptides detected in all three replicates is computed over the eight time points. Phosphopeptides that do not significantly respond to EGF stimulation can have low or negative correlation because the phosphorylation intensity exhibits minor, noisy fluctuations around the baseline level over time.

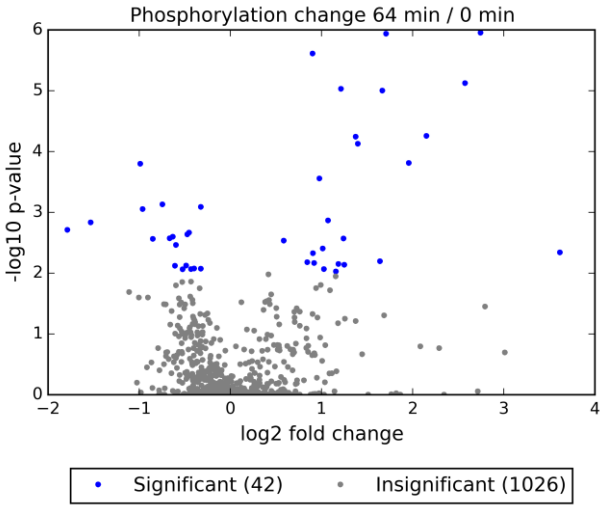

Phosphorylation change 2 min / 0 min
-log10 p-value
log2 fold change
Significant (99)
Insignificant (969)

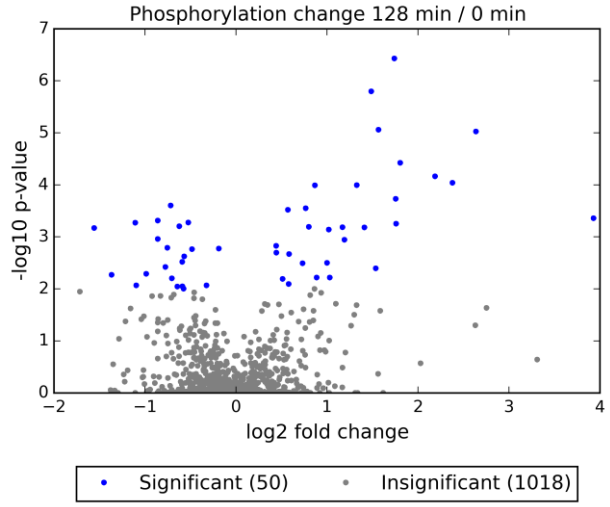

Phosphorylation change 4 min / 0 min
-log10 p-value
log2 fold change
Significant (131)
Insignificant (937)

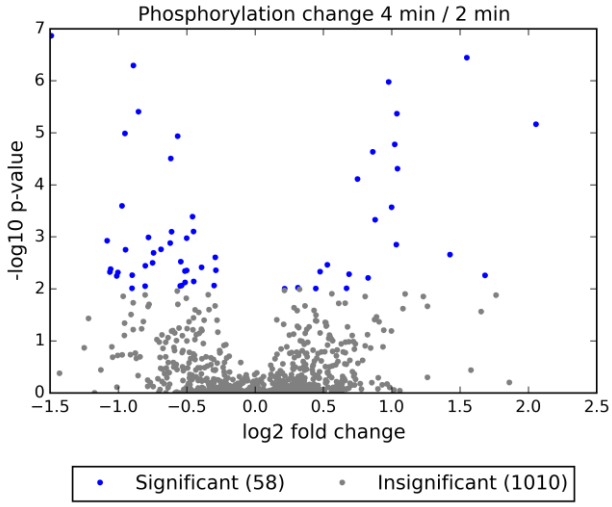

Phosphorylation change 8 min / 0 min
-log10 p-value
log2 fold change
Significant (95)
Insignificant (973)

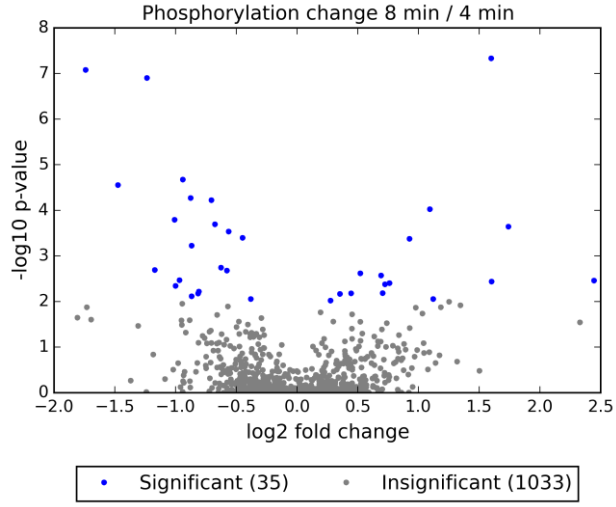

Phosphorylation change 16 min / 0 min
-log10 p-value
log2 fold change
Significant (80)
Insignificant (988)

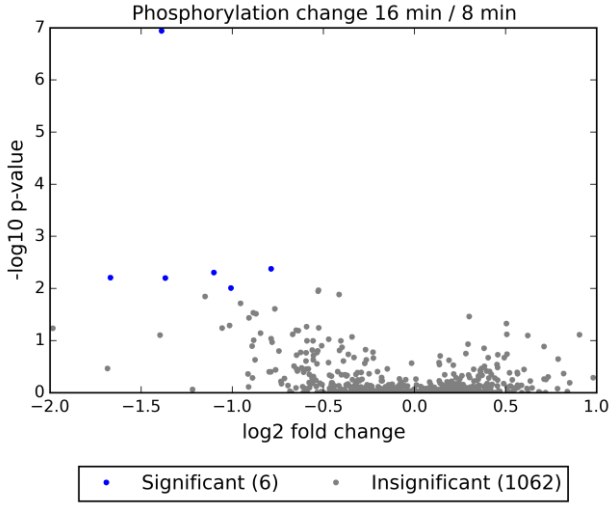

Phosphorylation change 32 min / 0 min
-log10 p-value
log2 fold change
Significant (54)
Insignificant (1014)

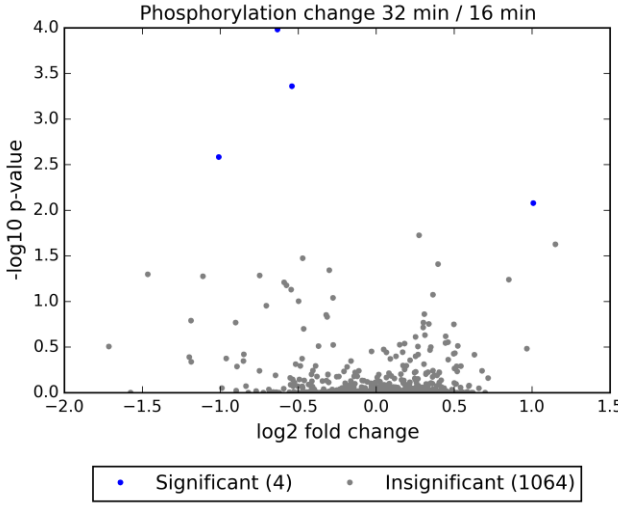

Phosphorylation change 64 min / 0 min
-log10 p-value
log2 fold change
Significant (42)
Insignificant (1026)

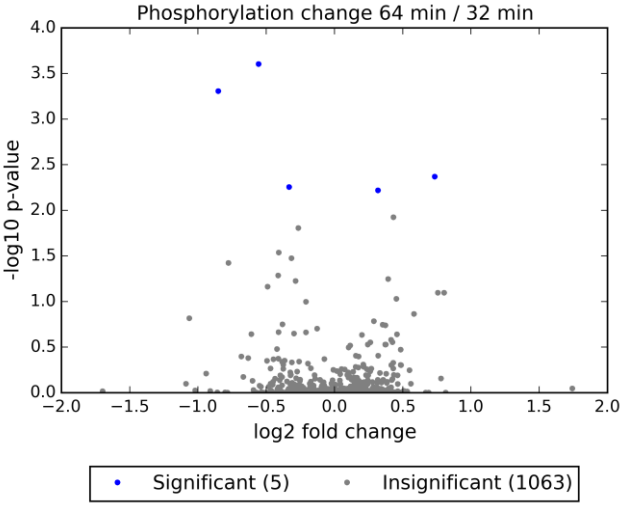

Phosphorylation change 128 min / 0 min
-log10 p-value
log2 fold change
Significant (50)
Insignificant (1018)


Phosphorylation change 4 min / 2 min
-log10 p-value
log2 fold change
Significant (58)
Insignificant (1010)


Phosphorylation change 8 min / 4 min
-log10 p-value
log2 fold change
Significant (35)
Insignificant (1033)


Phosphorylation change 16 min / 8 min
-log10 p-value
log2 fold change
Significant (6)
Insignificant (1062)


Phosphorylation change 32 min / 16 min
-log10 p-value
log2 fold change
Significant (4)
Insignificant (1064)


Phosphorylation change 64 min / 32 min
-log10 p-value
log2 fold change
Significant (5)
Insignificant (1063)

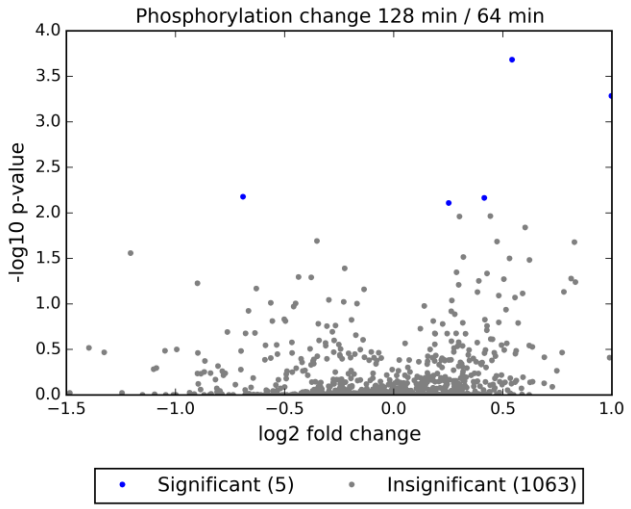


Figure 4. Volcano plots compare the statistical significance from Tukey's HSD test and the $\log_2$ fold change of the median phosphorylation intensity for all 1,068 phosphopeptides that were observed in all three biological replicates. When two of the three intensities were zero, the single non-zero intensity was used instead of the median to avoid dividing by zero.

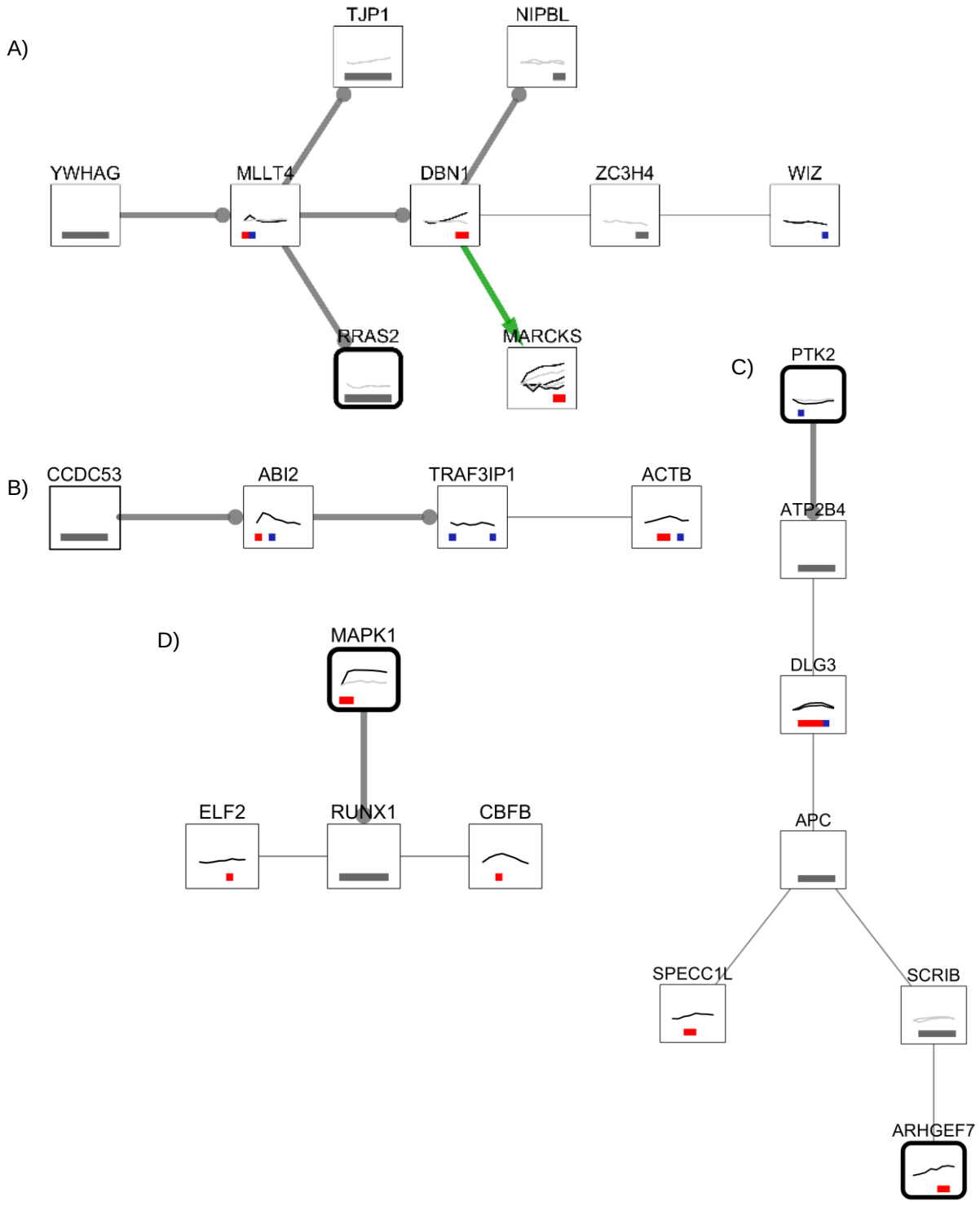
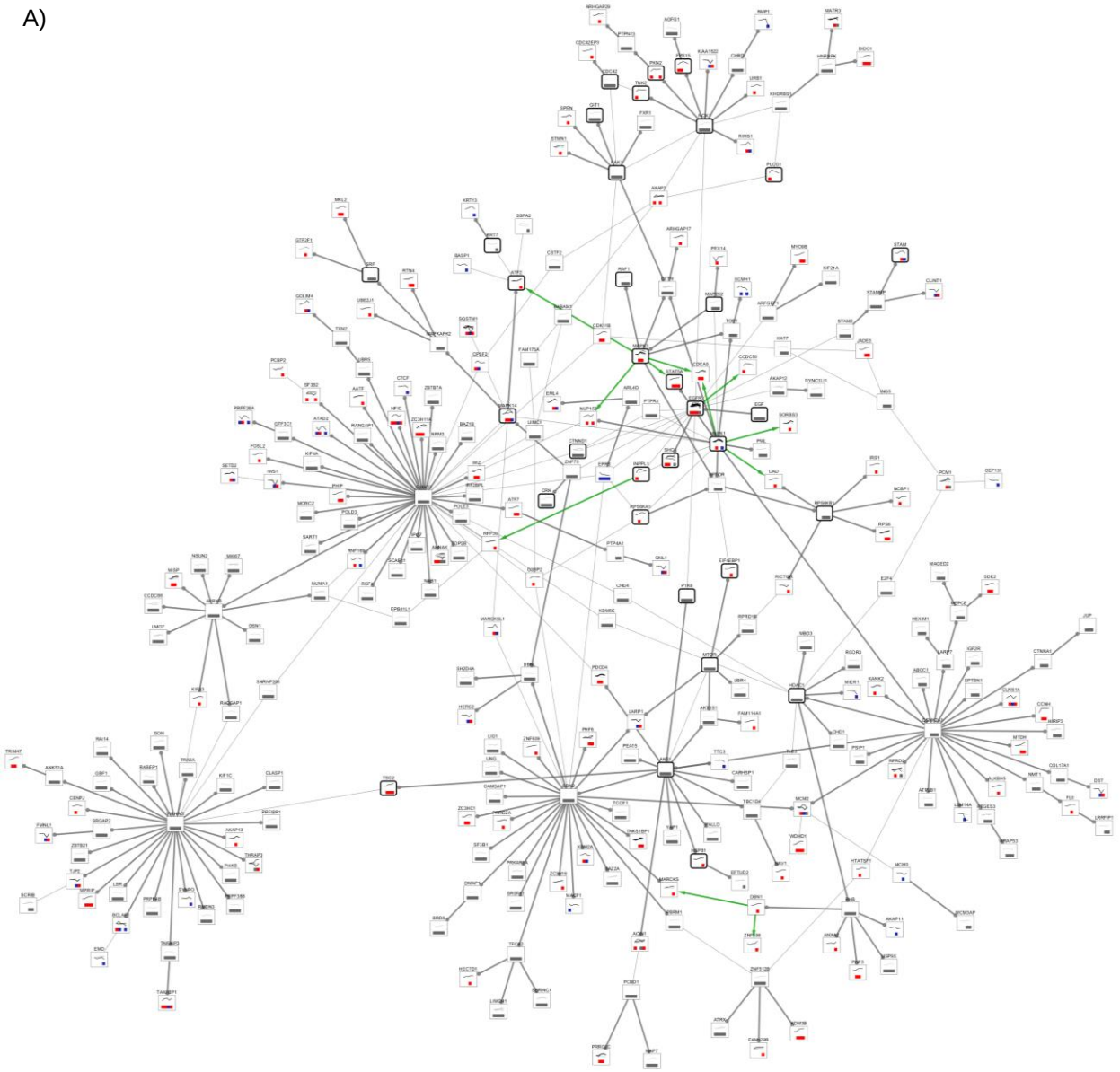


Figure 5. A) The subnetwork proximal to DBN1, MLLT4, and ZC3H4 shows how temporal constraints can propagate through the network to influence the direction of edges in other parts

of the pathway. Green edges depict activation. Gray edges that terminate in a circle indicate that the interaction is used in the same direction in all possible pathway models, but the sign is ambiguous. Thin, undirected edges are used in different directions in different valid pathway models. Thick, rounded borders show which proteins are present in one or more reference EGFR pathways. Node visualize phosphorylation fold changes and predicted protein activity. B) The ABI2 subnetwork (ABI2, TRAF3IP1, and their neighbors) shows how temporal constraints on one edge (TRAF3IP1-ACTB) can influence the orientation of different edges (the two directed edges involving ABI2) when performing global inference in the pathway model. C) The PTK2 subnetwork (encompassing all proteins on the path from ATP2B4 to SCRIB and their neighbors) demonstrates that SPECC1L and ARHGEF7 serve as temporal bottlenecks for paths from the source to PTK2 (these paths not shown). These proteins are activated later than PTK2, which implies that ATP2B4 cannot inhibit PTK2 at 4 min, the time it first responds to stimulation. D) RUNX1 is not observed in the phosphorylation data, but the timing of its neighbors' phosphorylation reveals that RUNX1 must be downstream of MAPK1.

A)



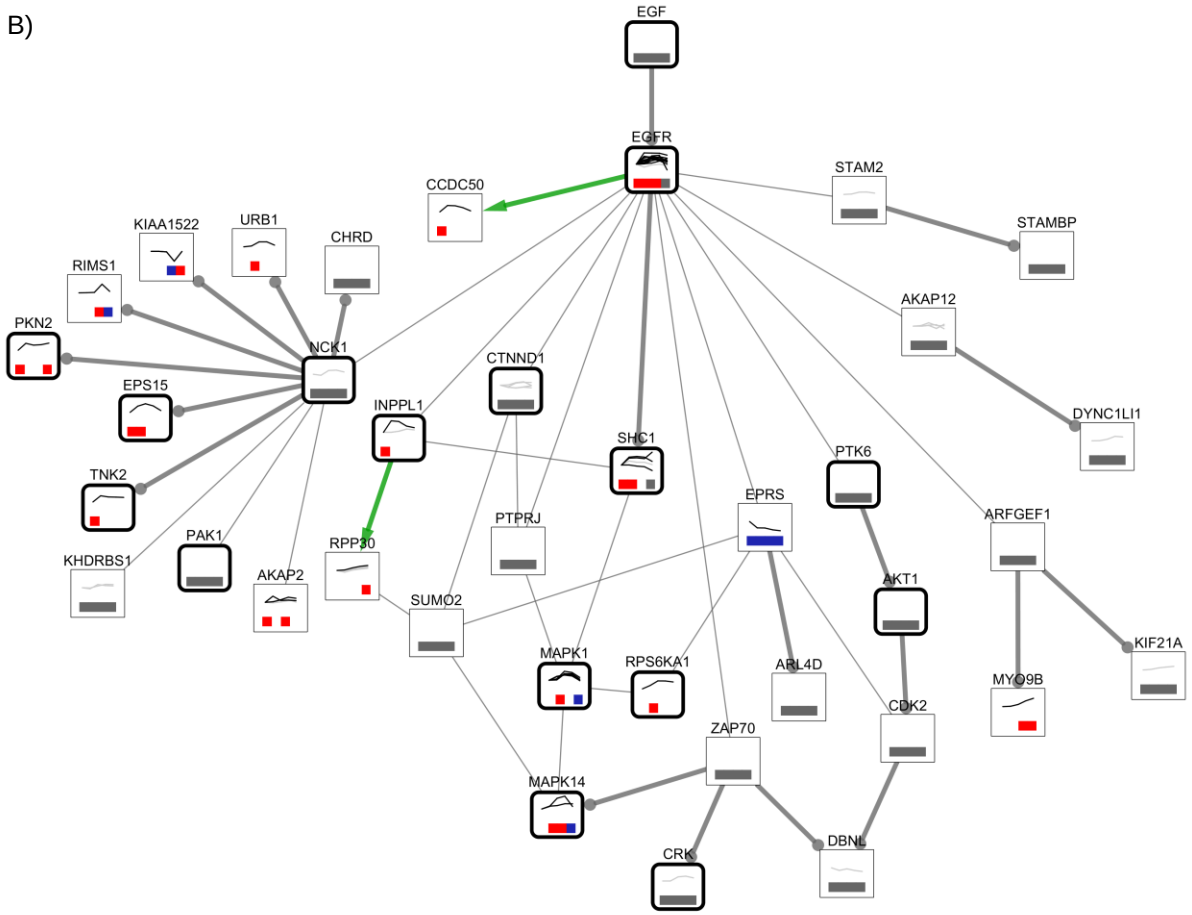


Figure 6. A) The full Temporal Pathway Synthesizer (TPS) pathway summary for the Olsen et al. EGF response in HeLa cells. B) A zoomed view of the same TPS pathway depicting EGFR and all proteins that are one or two interactions away in the TPS model.

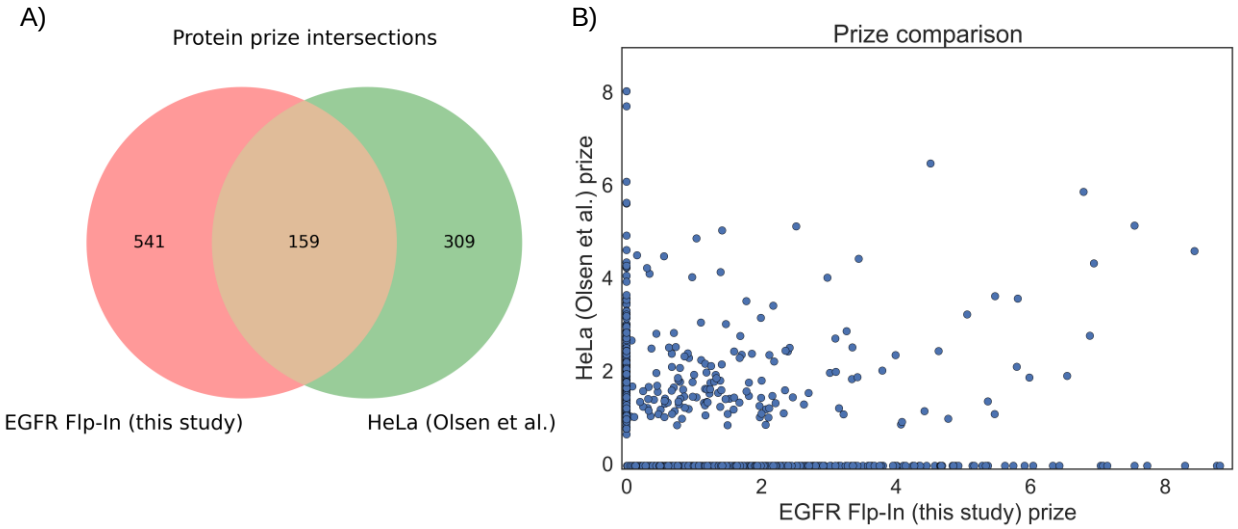


Figure 7. A) The set of proteins with quantified phosphorylation changes differ substantially in our study and the Olsen et al. study. The Venn diagram depicts only proteins that can be mapped to a UniProt identifier. B) Differences in the 1,009 protein prizes, the node importance scores used to select a protein-protein interaction subnetwork with the prize-collecting Steiner forest (PCSF) algorithm, lead to the disparate subnetworks. These prizes do not reflect the sign of the phosphorylation change. Proteins that have no quantified phosphopeptides in one of the studies receive a default prize of zero.

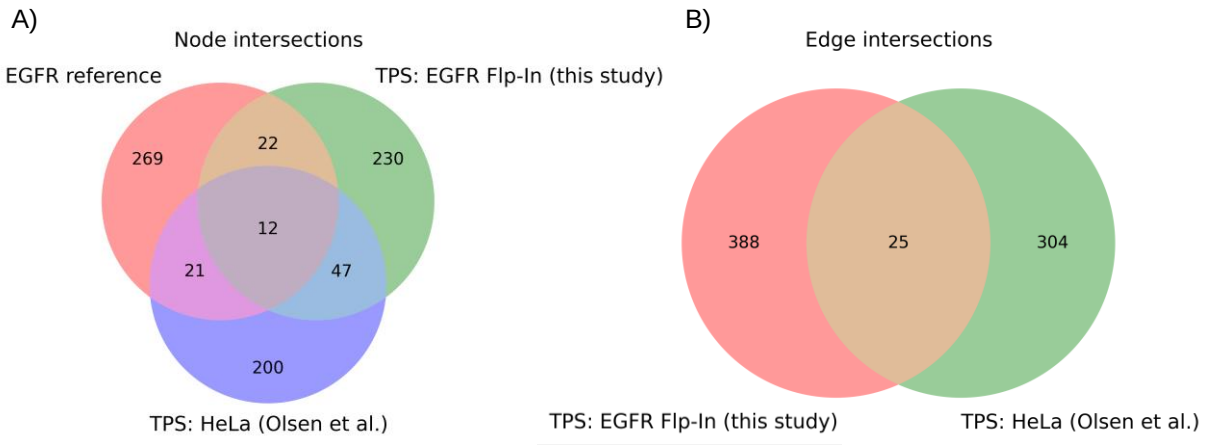
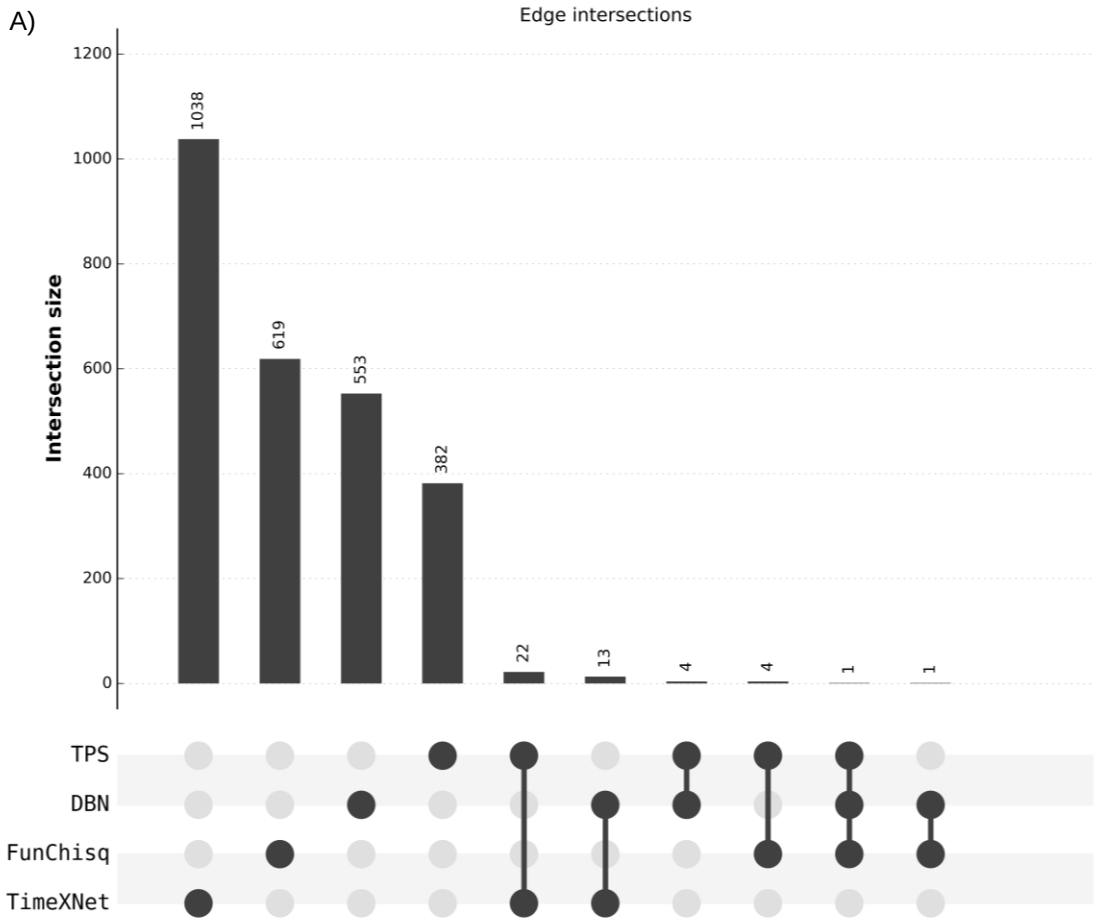


Figure 8. A) The protein overlap in the EGFR Flp-In and HeLA cell EGF response TPS pathway models and their recovery of EGFR reference pathway members. B) The edge overlap of the two EGF response TPS pathway models.


A)
Edge intersections
Intersection size
1200
1000
800
600
400
200
0
1038
619
553
382
22
13
4
4
1
1
TPS
DBN
FunChisq
TimeXNet

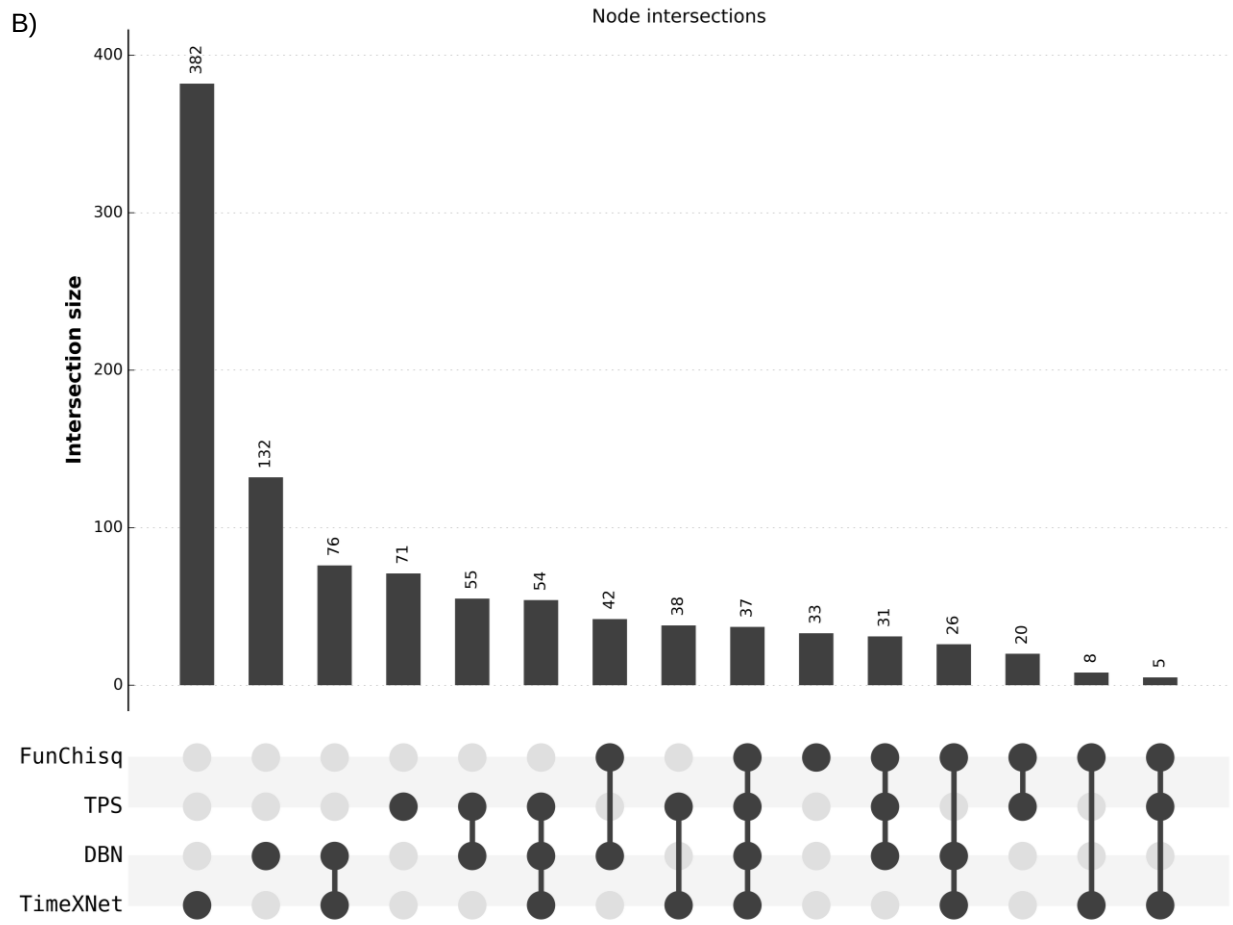


Figure 9. A) The edge overlap for the TimeXNet, Temporal Pathway Synthesizer, dynamic Bayesian network, and FunChisq predictions. The UpSet plot generalizes a Venn diagram. The circles depict subsets of networks, and the bars show the number of interactions predicted by that combination of networks. Directed and undirected interactions are treated as distinct, non-overlapping predictions. B) The node overlap for the four predicted networks.

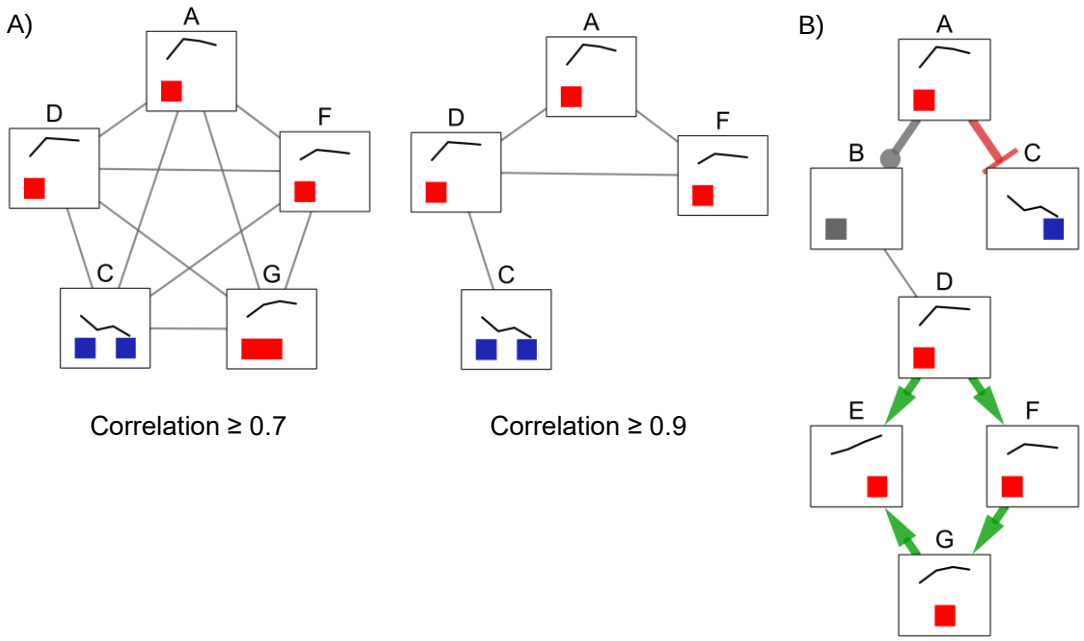


Figure 10. A) The networks obtained by thresholding the correlation of the phosphorylation profiles in the artificial example. Correlation is calculated as the absolute value of Pearson's correlation coefficient on the original data, not on the $\log_2$ fold changes used to visualize the phosphorylation changes over time in the network nodes. B) The network obtained by running TPS with all three types of constraints but restricting proteins to be active only at the time point when they achieve their peak phosphorylation change. This restriction induces errors in the predicted pathway.

Tables

Table 1. The number of edges that can be assigned: (1) a unique direction (Directed edges), and (2) a unique direction and sign (Signed, directed edges) across all valid pathway models for each combination of constraint type. In this study, "prior knowledge" refers to directed kinase-substrate interactions.

Constraints used	**Directed edges**	**Signed, directed edges**
Topology	119	0
Temporal	27	21
Prior knowledge	78	0
Topology + Temporal	148	32
Topology + Prior knowledge	182	0
Temporal + Prior knowledge	99	28
Topology + Temporal + Prior knowledge	202	38

Table 2. For each constraint type combination, we show: (1) "Added directed edges," or the number of edges that are uniquely directed when concurrently using all of the constraints listed in the row but not using any single constraint type, (2) "Added signed, directed edges," or the equivalent number for edges that are uniquely signed and directed, and (3) "Conflicts," or the number of edges allowed by at least one constraint type that conflict with another constraint type when the properties are concurrently asserted. Conflict edges are excluded from the final network.

Constraints used	**Added directed edges**	**Added signed, directed edges**	**Conflicts**
Topology + Temporal	8	12	9
Topology + Prior knowledge	2	0	0
Temporal + Prior knowledge	0	7	8
Topology + Temporal + Prior knowledge	7	18	9

Table 3. Aggregate statistics for 100 runs with a randomized peptide to protein mapping. We show 25th, 50th, and 75th percentile statistics for (1) unambiguously directed and (2) unambiguously signed and directed edges per category. "Unambiguous" edges are those that can only be assigned a unique direction (respectively, a unique direction and sign) across all directed graph models (respectively, signed, directed graph models). "Agreeing unambiguous" edges are those that are assigned the same unique direction (respectively, the same unique direction and sign) in the original and randomized network. "Conflicting unambiguous" edges are those that are assigned opposite unique directions (respectively, distinct sign and direction) in the original and randomized network. "Common prior knowledge" edges are those that appear in both the original and randomized network and that are also present in the prior knowledge (edges with known direction). Edges "only in the original network" are those that are assigned at least one direction and sign in the original network, but not in the randomized network. "Randomized networks with more edges" shows how many of the 100 runs have more directed edges and more signed and directed edges than the original network

Percentile	Edge type	Directed edges	Signed, directed edges
	Randomized networks with more edges	95	0
25	Unambiguous in the original network	202	38
	Unambiguous in randomized networks	252	14.5
	Agreeing unambiguous	25.5	0
	Conflicting unambiguous	0	0
	Common prior knowledge	13	0
	Only in the original network	166.5	37
	Only in randomized networks	223	14
50	Unambiguous in the original network	202	38
	Unambiguous in randomized networks	270	20
	Agreeing unambiguous	30.5	0
	Conflicting unambiguous	0	0
	Common prior knowledge	18	0
	Only in the original network	171.5	38
	Only in randomized networks	239	19.5
75	Unambiguous in the original network	202	38
	Unambiguous in randomized networks	286	26
	Agreeing unambiguous	35	0
	Conflicting unambiguous	1	0
	Common prior knowledge	22	0
	Only in the original network	176	38
	Only in randomized networks	253	26

Table 4. Aggregate statistics for 100 runs with randomized time points within each time course. The rows and columns are as described in Table 3.

Percentile	Edge type	Directed edges	Signed, directed edges
	Randomized networks with more edges	1	0
25	Unambiguous in the original network	202	38
	Unambiguous in randomized networks	189	21
	Agreeing unambiguous	178	14
	Conflicting unambiguous	2	1
	Common prior knowledge	77	4.5
	Only in the original network	18	16
	Only in randomized networks	8	4
50	Unambiguous in the original network	202	38
	Unambiguous in randomized networks	192	24
	Agreeing unambiguous	179	16
	Conflicting unambiguous	2	2
	Common prior knowledge	78	6.5
	Only in the original network	20	19
	Only in randomized networks	10	5
75	Unambiguous in the original network	202	38
	Unambiguous in randomized networks	195	28
	Agreeing unambiguous	181	20
	Conflicting unambiguous	3	3
	Common prior knowledge	78	8
	Only in the original network	22	22
	Only in randomized networks	12	7

Table 5. Aggregate statistics for 100 runs with a randomized PPI network. The rows and columns are as described in Table 3.

Percentile	Edge type	Directed edges	Signed, directed edges
	Randomized networks with more edges	91	57
25	Unambiguous in the original network	202	38
	Unambiguous in randomized networks	237	35
	Agreeing unambiguous	2	0
	Conflicting unambiguous	0	0
	Common prior knowledge	2	0
	Only in the original network	197	36
	Only in randomized networks	234	33.5
50	Unambiguous in the original network	202	38
	Unambiguous in randomized networks	251	40.5
	Agreeing unambiguous	3	1
	Conflicting unambiguous	0	0
	Common prior knowledge	3	1
	Only in the original network	198.5	37
	Only in randomized networks	248	40
75	Unambiguous in the original network	202	38
	Unambiguous in randomized networks	270	46
	Agreeing unambiguous	5	2
	Conflicting unambiguous	0	0
	Common prior knowledge	5	1
	Only in the original network	200	38
	Only in randomized networks	267	45

Table 6. Aggregate statistics for 100 bootstrap runs with 10, 25, or 50% of the significant phosphopeptides randomly removed. The rows and columns are as described in Table 3.

Removed	Percentile	Edge type	Directed edges	Signed, directed edges
10		Randomized networks with more edges	28	9
	25	Unambiguous in the original network	202	38
		Unambiguous in randomized networks	188	31.5
		Agreeing unambiguous	148	28
		Conflicting unambiguous	0	0
		Common prior knowledge	54	8
		Only in the original network	41	5
		Only in randomized networks	31	2
	50	Unambiguous in the original network	202	38
		Unambiguous in randomized networks	196.5	34
		Agreeing unambiguous	154	32
		Conflicting unambiguous	0	0
		Common prior knowledge	59	9
		Only in the original network	48	6
		Only in randomized networks	41	3
	75	Unambiguous in the original network	202	38
		Unambiguous in randomized networks	203	37
		Agreeing unambiguous	161	33
		Conflicting unambiguous	0	0
		Common prior knowledge	64	10
		Only in the original network	54	10
		Only in randomized networks	51	4
25		Randomized networks with more edges	0	0
	25	Unambiguous in the original network	202	38
		Unambiguous in randomized networks	160	25
		Agreeing unambiguous	101	20
		Conflicting unambiguous	0	0
		Common prior knowledge	38	6
		Only in the original network	83	12
		Only in randomized networks	49.5	4

	50	Unambiguous in the original network	202	38
		Unambiguous in randomized networks	169	29
		Agreeing unambiguous	110.5	23
		Conflicting unambiguous	0	0
		Common prior knowledge	44	7
		Only in the original network	91	15
		Only in randomized networks	60	5
	75	Unambiguous in the original network	202	38
		Unambiguous in randomized networks	182	31
		Agreeing unambiguous	119	26
		Conflicting unambiguous	0	0
		Common prior knowledge	50	8
		Only in the original network	101	18
		Only in randomized networks	70	7
50		Randomized networks with more edges	0	0
	25	Unambiguous in the original network	202	38
		Unambiguous in randomized networks	108	13
		Agreeing unambiguous	50.5	7
		Conflicting unambiguous	0	0
		Common prior knowledge	21.5	2
		Only in the original network	138	26
		Only in randomized networks	52	4
	50	Unambiguous in the original network	202	38
		Unambiguous in randomized networks	119	15
		Agreeing unambiguous	59.5	9
		Conflicting unambiguous	0	0
		Common prior knowledge	26	4
		Only in the original network	142.5	29
		Only in randomized networks	58.5	6
	75	Unambiguous in the original network	202	38
		Unambiguous in randomized networks	127	20
		Agreeing unambiguous	64	12
		Conflicting unambiguous	0	0
		Common prior knowledge	30	5
		Only in the original network	151	31
		Only in randomized networks	67	9

Table 7. Temporal activity distributions in the mass spectrometry phosphoproteomic data and the TPS pathway model. Significant peptides are those whose phosphorylation significantly changes with respect to the previous time point or baseline level, and the percentage is relative to all peptides that are significant at any time point (n = 263). The pathway proteins are all proteins in the TPS pathway summary that are active at each time point, and the percentage is relative to all pathway members (n = 311). Temporally-constrained proteins are the subset of pathway proteins that are associated with a significantly changing peptide (n = 168).

Time (min)	**Significant peptides**	**TPS pathway proteins**	**Temporally-constrained proteins**
2	99 (38%)	189 (61%)	58 (35%)
4	153 (58%)	222 (71%)	82 (49%)
8	115 (44%)	197 (63%)	56 (33%)
16	86 (33%)	169 (54%)	28 (17%)
32	56 (21%)	166 (53%)	25 (15%)
64	45 (17%)	160 (51%)	17 (10%)
128	52 (20%)	165 (53%)	22 (13%)

Table 8. The temporal distributions of the times when a peptide is first significant or a protein is first active. Peptide and protein categories are as described in Table 7. The temporally-constrained percentages sum to 101% due to rounding to whole percentages.

Time (min)	Significant peptides	TPS pathway proteins	Temporally-constrained proteins
2	99 (38%)	189 (61%)	58 (35%)
4	84 (32%)	66 (21%)	57 (34%)
8	32 (12%)	22 (7%)	21 (13%)
16	12 (5%)	4 (1%)	4 (2%)
32	11 (4%)	10 (3%)	10 (6%)
64	8 (3%)	8 (3%)	6 (4%)
128	17 (6%)	12 (4%)	12 (7%)

Table 9. The number of unambiguous predictions made for different values of the time course significance p-value threshold. In the "directed" analysis, an edge is unambiguous if it is assigned a unique direction. In the "signed, directed" analysis, an edge is unambiguous if it is assigned a unique sign and direction. "Common edges" are the number of unambiguous edges that are in common with the TPS model obtained with the default choice of p-value, 0.01, which has 202 unambiguous directed edges and 38 unambiguous signed directed edges.

Analysis	**p-value**	**Unambiguous edges**	**Common edges**
Directed	0.001	203	190
	0.005	204	194
	0.05	213	192
	0.1	199	178
Signed, directed	0.001	22	16
	0.005	28	26
	0.05	49	27
	0.1	38	15

Results

TPS EGFR pathway model recapitulates known biology

The TPS pathway model includes 41 kinases and 5 phosphatases as well as adaptors and other types of proteins that coordinate with the direct phosphorylation regulators. Some of the differentially phosphorylated proteins in the TPS model may not be functional, but the TPS network provides a framework to study their role in the EGF response. In addition to the phosphorylated proteins, 38 other proteins are included in the signaling pathway as hidden intermediate nodes that propagate signals via different mechanisms.

Four sites on EGFR are immediately and significantly phosphorylated post-stimulation as are two others at later time points. Pathway models could potentially begin with EGF interacting with any of its 17 partners in the PPI network, but strong EGFR phosphorylation leads TPS to initiate all paths with the edge EGF->EGFR. Proteins directly connected to EGFR include known pathway members CBL, CBLB, EPN1, HGS, GRB2, PAK1, PRKACA, and PTK2. SHC1 is connected to EGFR via GRB2, but the direction and sign of the interaction are ambiguous due to a second parallel connection through CBL, which is also active at the two- and four-minute time points. Likewise, PLCG1 is connected via CBL, again with indeterminate direction and sign. This reflects how quickly EGFR and the other upstream pathway members are activated, suggesting that sub-minute time points may be required to unambiguously order some of the immediate connections adjacent to EGFR (Reddy et al., 2016).

Further downstream, MAP2K1 is one of several canonical EGFR pathway members that are not phosphorylated in our mass spectrometry data but are included in the pathway. Such proteins emphasize the necessity of including PPI in the analysis of the temporal phosphorylation changes because these unobserved proteins could not be recovered by any algorithm that reconstructs the pathway from the mass spectrometry data alone. MAP2K1 is correctly recognized as the direct kinase of EGFR pathway members MAPK1 (after adding new experimental constraints described below) and MAPK3. MAPK1 and MAPK3 phosphorylation levels are highly correlated and would likely be directly linked by an approach based on correlation or mutual information, but TPS correctly predicts that MAPK1 and MAPK3 correlation is due to the common upstream regulator (MAP2K1) instead. Immediately downstream of these proteins, MKL1 phosphorylation is not as strongly correlated as the two MAPKs, but TPS combines the topological constraints with the temporal information to correctly recover MAPK1->MKL1 and MAPK3->MKL1 (Muehlich et al., 2008).

The most compelling evidence supporting the predicted edge directions comes from the EGFR reference pathways, which confirm both the edge direction and relevance to EGF stimulation response. Seven directed edges appear in one of the reference pathways, four with the predicted direction and three in complexes. We expect this overlap to be low because so few significantly phosphorylated proteins are in the reference pathways. In addition, 78 directed edges come from the PhosphoSitePlus input data (Hornbeck et al., 2015), in which the kinase-substrate interaction direction is already known to be correct in other contexts. Nearly all of

these interactions were not previously reported to be involved in EGFR signaling; only three are present in an EGFR reference pathway.

Natural language processing (NLP) tools confirm edge directions

The NLP software systematically searches literature and eliminates many of the biases present in manual literature searches, but the extracted evidence can contain errors. Therefore, we use three NLP tools – iHOP (Hoffmann and Valencia, 2004), Chilibot (Chen and Sharp, 2004), and Literome (Poon et al., 2014) – to identify PubMed abstracts that describe the predicted interactions and then manually review each abstract to score the degree of support for the prediction. We focus on 54 edges in the TPS network that are on the periphery of the EGFR reference pathways because these predictions directly extend the known pathway and are likely to be relevant to EGF response. Of these edges, 16 have a specific direction described in the abstracts retrieved via NLP, and 15 of those directions are predicted correctly. EGFR regulating CCDC50 (also known as Ymer) is the only incorrect edge in the NLP evaluation (Tashiro et al., 2006). However, PhosphoSitePlus indicates EGFR is a putative CCDC50 kinase because EGFR inhibition suppresses CCDC50 phosphorylation (Kameda et al., 2009), supporting that prediction as well.

Combining multiple constraints reduces pathway ambiguity

We performed a comparative analysis of networks inferred using different subsets of constraints in order to: (1) quantify the individual contribution of different kinds of constraints in our joint inference procedure, (2) show that more predictions can be made as we combine

different types of data and constraints, and (3) rule out predictions that are supported by a single type of data but in conflict with other types.

Each type of constraint restricts the space of valid models in different ways and leads to different (but possibly overlapping) sets of inferred pathway edges. We compare constraint types and their combinations in terms of the set of interactions that can be assigned a unique direction (and sign) across all models. For each combination of constraints, we calculate how many interactions can be uniquely directed or uniquely signed and directed (Table 1). This analysis reveals the impact of different constraint types on valid pathway models. For instance, the topological constraints imposed by the input graph structure greatly contribute to the ability of TPS to infer directed edges. Meanwhile, temporal constraints are required to recover signed, directed edges. We also observe that as more constraints are added, the number of unambiguous directed (or signed, directed) interactions we recover increases. For many edges, using only one or two constraint types produces a collection of valid pathway models that conflict with respect to the direction or sign of the edge, but adding another constraint class lets TPS eliminate some of those models and make a definitive prediction. To better explain the interplay among combinations of constraints, we perform an additional analysis to identify edge predictions that can be made only if constraints are used together and conflicts that can be detected through joint reasoning (Table 2).

To illustrate how the combined topological and temporal constraints can exert nontrivial, non-local effects on the space of valid signed, directed networks, consider the following example. The path between YWHAG and MLLT4 cannot be oriented using any of the three constraint

types alone. By joining temporal and topological properties, TPS discovers that the only way to reach YWHAG from MLLT4 goes through DBN1, which is activated later (32-128 minutes) than MLLT4 (0-4 minutes). Therefore, no network contains an edge from MLLT4 to YWHAG, and the interaction can be uniquely oriented from YWHAG to MLLT4 (Figure 5A).

A complex case of non-local temporal effects is observed around ABI2 (Figure 5B). With either topological or temporal constraints, it is not possible to orient the two edges incident to ABI2 that link it to CCDC53 and to TRAF3IP1. Reasoning using both types of constraints, TPS narrows down the possible orientations to CCDC53->ABI2->TRAF3IP1. Making this inference requires looking at the temporal data for ABI2, TRAF3IP1, and TRAF3IP1's neighbor ACTB. ABI2 can be active at 0-2 or 4-8 minutes, TRAF3IP1 can be active at 0-2 or 64-128 minutes, and ACTB can be active at 4-16 or 32-64 minutes. As a result, having an edge ACTB->TRAF3IP1 implies that TRAF3IP1 is active late, and it cannot precede the activity of ABI2. There is therefore no tree network that has an edge from TRAF3IP1 to ABI2 or an edge from ABI2 to CCDC53.

The direction prediction from PTK2 to ATP2B4 (Figure 5C) is inferred in two independent ways. This direction is given in the curated kinase-substrate interactions, but is also a logical consequence of combining the tree topology with time series data. The latter inference is obtained by reasoning on multiple alternative paths at once. ATP2B4 cannot precede PTK2, because all paths from the source to ATP2B4 that do not go through PTK2 must go through either SPECC1L or ARHGEF7. Both of these proteins are active strictly later than PTK2, therefore they cannot precede PTK2 in any valid model.

The same phenomenon, namely a direction prediction through two independent mechanisms, is manifested in the MAPK1 to RUNX1 interaction (Figure 5D). This interaction direction is given by the kinase-substrate interactions, but the direction is independently inferred by looking at the time series data in a non-local fashion. Even though RUNX1 does not have any phosphorylation data, it cannot precede MAPK1 in any tree model because its other neighbors, ELF2 and CBFB, succeed MAPK1 in their activation.

Permuting the phosphoproteomic and network data changes TPS predictions

In addition to assessing the impact that asserting temporal constraints has on TPS predictions, we considered the effect of permuting the time course and PPI network data on the TPS output network. We performed three different randomization strategies with our EGF response dataset: (1) permuting the peptide to protein mapping, (2) permuting the time point order within each peptide time course, and permuting the background PPI network (Tables 3–5). Finally, we carried out a bootstrap analysis by holding out 10%, 25%, or 50% of significant phosphopeptides (Table 6).

In the case of the permuted peptide to protein mapping, we observe that a majority of predictions made using randomized data are disjoint from those inferred from the original data (Table 3). The median number of edges that are unambiguously directed using the original data but are missing from the randomized runs is 171.5 out of a total of 202 predictions. The difference between signed, directed predictions is even more pronounced. The median number of original unambiguous predictions that are missing from the randomized runs is all 38 out of 38, and no randomized run has more signed, directed predictions than the original output.

Meanwhile, a comparison between the original and permuted data with respect to reference pathways is inconclusive. The impact of temporal constraints on top of topological and prior knowledge constraints is relatively small, and the edges that can be checked against reference pathways is often shared between the randomized and non-randomized cases due to the shared prior knowledge of kinase-substrate directions.

Permuting the phosphopeptides' temporal information has less impact than permuting the peptide to protein mapping (Table 4). Individual phosphopeptides have their own temporal activities, but the interaction network is at the protein level instead of the peptide level. Therefore, there are permutations in which some permuted phosphopeptide on a protein is active as early as the earliest phosphopeptide in the real data and still precedes the activation or inhibition of its neighboring proteins. This is more likely for proteins with many quantified phosphopeptides. As more site-specific kinase-substrate and phosphatase-substrate interactions become available, site-specific interaction networks can reduce this ambiguity without modifications to the TPS algorithm. In addition, the prior knowledge and topological constraints substantially influence the pathway signs and directions and are not affected by the temporal permutation. Nevertheless, TPS does make more unambiguous predictions with the original data compared to the permuted data, especially in the case of signed, directed predictions.

When the PPI network is permuted, TPS makes a large number of unambiguous predictions, but the overlap of these predictions with the ones made using non-randomized data is very low

(Table 5). Furthermore, there is low overlap between predictions made using randomized PPI networks and the reference pathways.

In our bootstrap analysis, we note that TPS makes more than half the amount of directed predictions when only 50% of the significant phosphopeptides are used (Table 6). We also observe that no conflicts with the original TPS output network are introduced in the bootstrap analysis.

Signaling timing

Pathway activity peaks at four minutes, with a substantial decrease in the number of active proteins at the late time points (Table 7). The pathway contains 143 proteins that are either Steiner nodes identified by PCSF (not associated with any peptide in the mass spectrometry data) or only associated with peptides whose phosphorylation does not significantly change. These proteins could be active at any time point *a priori*, but TPS is able to use the topological relationships with other proteins to infer more specific temporal activity windows for 12 (8%). The time at which pathway members are first active reveals how rapidly signals propagate from EGF (Table 8). More proteins are first active at two minutes post-stimulation than any other time, and within eight minutes almost all proteins on the pathway respond to the stimulus.

Even when normalizing for the number of significantly differentially phosphorylated proteins at each time point, the EGF-responsive proteins have the greatest overlap with the EGFR reference pathways at the initial 2 min time point. For some of the EGFR reference pathways, there are comparable overlaps at later time points. For others, the 2 min overlap is by far the greatest. When considering the union of all BioCarta, Reactome, or PID pathways, however, the

later time points do exhibit overlaps that are similar to or greater than the 2 min overlap. Babur et al. provide additional visualizations of the temporal pathway activity of our EGF response data (Babur et al., 2018).

TPS generates high-quality pathway models from lower temporal resolution phosphoproteomic spectrometry data

Existing EGF response temporal phosphoproteomic datasets covered fewer time points than our dataset (Olsen et al., 2006) or were limited to tyrosine phosphorylation (Oyama et al., 2009; Zhang et al., 2005). A more recent study (Reddy et al., 2016) complements ours, providing a high-temporal-resolution view of the early signaling dynamics but covering a smaller fraction of the overall EGFR pathway. Having established that TPS produces informative signaling pathway models from our EGF stimulation phosphorylation data, we assessed whether it could attain similar performance on a dataset with lower temporal resolution.

The foundational study by Olsen et al. (Olsen et al., 2006) examined the phosphorylation response to EGF at 0, 1, 5, 10 and 20 minutes post-stimulation. We identified 302 phosphopeptides on 203 proteins with a phosphorylation $\log_2$ fold change of at least 2 and constructed PCSF and TPS pathway models from these proteins that exhibited temporal phosphorylation changes. The resulting TPS model connects 280 proteins with 329 edges, and 242 edges can be assigned a unique direction in all possible pathway structures (Figure 6). Twenty-four (10%) of the 243 phosphorylated proteins and 9 (24%) of the 37 unphosphorylated connective proteins are known EGFR pathway members. We evaluated the 242 edges assigned a unique direction in the TPS model against the EGFR-specific and generic reference pathways.

The median number of directed edges that matched a directed edge in a reference pathway was two, and none of the reference pathways reported an interaction that conflicted with any of the directed predictions. Overall, TPS performs well even with fewer time points and differentially phosphorylated proteins.

Although the TPS pathways produced with our phosphoproteomic data and the Olsen et al. data are of similar sizes and contain an almost identical number of proteins from the EGFR reference pathways, a closer examination reveals that their pathway structures reflect the differences in the underlying phosphorylation data. The Olsen et al. EGF stimulation study used HeLa cells, a shorter time course, and different phosphorylation quantification techniques than our approach and did not provide biological replicates. This led to substantial differences in the quantified proteins and the magnitudes of their phosphorylation changes (Figure 7), which consequently have a strong influence on the resulting network models. Only 59 proteins and 25 edges are common to both TPS models, and each TPS EGFR network captures unique parts of the reference EGFR pathways (Figure 8). Each network reflects the individual aspects of EGF signaling in the cell types and conditions being studied, though it is likely that the quantified phosphosites and generated TPS models would grow more similar with deeper phosphoproteome coverage.

Selecting interactions for Western blotting

To select edges for Western blotting from the 413 predicted interactions in the TPS EGFR pathway model, we first considered the 202 edges that were assigned a confident interaction direction. Next, we focused on the subset of 52 edges that extended the canonical EGFR

pathway models. These directed edges contained one protein that is a member of a reference pathway and one that is not. Finally, we prioritized the 36 edges for which there was a known kinase-substrate interaction or NLP support. Thus, these directed interactions are not known to be involved in the EGFR pathway but do have prior evidence in different cellular contexts. From these 36 edges, we selected 9 to test based on inhibitor and antibody availability. We added a tenth edge, MAPK1-ATP1A1, that had a confident edge sign but an ambiguous direction because it extended the canonical EGFR pathway and had a high-quality inhibitor and antibody available.

We initially tested inhibitor and antibody combinations with a limited number of replicates and then performed additional replicates if the band and phosphorylation change could be detected reliably. In six cases, no band was detected. RANBP3 exhibited moderate changes in phosphorylation in response to AKT1 inhibition. However, these changes were not as substantial as those observed on CRK, ATP1A1, or ZYX so we prioritized those for further replicates. MAP2K1 phosphorylation was not affected by CDK1 inhibition. This suggests that the TPS prediction CDK1->MAP2K1 is incorrect, a different MAP2K1 phosphorylation site is impacted, or the RO-3306 compound did not penetrate the cell membrane.

Iterative experimental and computational modeling further reduces pathway ambiguity

An important feature of TPS is its flexibility to integrate different types of constraints on pathway structure. This makes it ideal for iterative modeling because computational hypotheses that are experimentally confirmed or refuted can be fed back into TPS. Based on the

results from the Western blots, we added a new constraint: MAPK1 inhibits ATP1A1. We then ran TPS again, requiring pathways to be consistent with the new constraint and all previous constraints. After restricting the pathway structure, TPS correctly infers that MAP2K1 is directly upstream of MAPK1 yielding a more precise and accurate pathway. Without the Western blot-derived constraint, the direction of the MAP2K1-MAPK1 interaction was ambiguous due to the possibility that ATP1A1, rather than MAP2K1, controls MAPK1 phosphorylation changes. Other types of experimental corroboration can be similarly applied to iteratively improve the predictive power of TPS.

Role of tested EGFR pathway edges

The EGFR pathway provides a rich background against which the TPS predictions can be evaluated, while concurrently letting us evaluate its ability to uncover new biology in an extensively studied and clinically relevant pathway. EGFR initiates a variety of signaling cascades that regulate phenotypes such as cell differentiation, migration, proliferation, and survival (Citri and Yarden, 2006). Dysregulation of EGFR activity by overexpression, mutation, or other mechanisms leads to cellular disease, including cancer (Huang et al., 2009; Pines et al., 2010).

All three experimentally-tested pathway edges naturally extend traditional EGFR pathway representations. These interactions involve kinases that play recognized roles in EGF response and introduce connections to target proteins that are controlled by the kinases in the EGFR context. We know *a priori* from the PPI network that the proteins physically interact in some

manner. TPS illuminates the interactions' relevance to EGF response and the details of the interactions, that is, the directions and signs.

ABL2, a well-characterized tyrosine kinase also known as ARG, is involved in actin remodeling, cell motility, and EGFR endocytosis (Colicelli, 2010). Though not detected in our mass spectrometry data, this protein is included in the TPS pathway due to its predicted role as a regulator of CRK, which is highly phosphorylated in response to EGF. ABL2 has been shown to phosphorylate CRK on Y221 *in vivo* (Wang et al., 1996). We confirm this site is sensitive to Dasatinib, but it differs from the CRK site detected in our mass spectrometry data (Y136). KEGG's ErbB signaling pathway depicts CRK as a direct regulator of ABL2, which contradicts our evidence that ABL2 directly phosphorylates CRK in the EGFR pathway. Both directions are potentially valid, but the current version of TPS does not model cycles.

The kinase MAPK1 (ERK2) is a central component of many stimulus responses and other biological processes whose dysfunction is linked with numerous diseases (Wortzel and Seger, 2011). ATP1A1 is the catalytic subunit of Na^+,K^+-ATPase, an ion pump that also plays a role in signal transduction when inhibited by ouabain (Reinhard et al., 2013), making it a candidate therapeutic target for diseases such as medulloblastoma (Wolle et al., 2014). MAPK1 phosphorylates ATP1A1 in response to insulin (Al-Khalili et al., 2004) and proinsulin-connecting peptide (C-peptide) (Zhong et al., 2004) stimulation, potentially on T81. However, both the TPS pathway model and our MAPK1 inhibition support an inhibitory effect on ATP1A1 phosphorylation. The difference could be attributed to the specific phosphorylation sites because MAPK1 was not linked to ATP1A1 tyrosine phosphorylation in other conditions

(Al-Khalili et al., 2004). ATP1A1 Y55 (insignificant change) and Y260 (significant change) exhibit decreased phosphorylation in our mass spectrometry data. Our Western blot quantifies phosphorylation of Y10, which was detected jointly with Y260 phosphorylation in many PhosphoSitePlus Cell Signaling Technology curation sets (Hornbeck et al., 2015). Despite prior evidence of the MAPK1-ATP1A1 interaction, we cannot rule out indirect effects via other inhibited proteins (SCH772984 also inhibits MAPK3) or an intermediate tyrosine phosphatase controlled by MAPK1, which could explain the significant change in ATP1A1 phosphorylation at 0 min.

AKT1 is an important kinase in PI3 kinase signaling with roles in glucose transport, cell survival, cell growth, metabolism, and multiple diseases (Hers et al., 2011). Like ABL2, AKT1 phosphorylation was not detected in our mass spectrometry data, but it is correctly nominated as a regulator of other EGF-responsive proteins. AKT1 directly phosphorylates Zyxin on S142 (Chan et al., 2007), an activating interaction. Our Western blots support AKT1 inhibiting Zyxin phosphorylation based on an antibody specific to S142 and S143. The disagreement in edge sign is an intriguing topic for further study. It may suggest that although AKT1 directly phosphorylates Zyxin some contexts, their relationship in the EGF response context is indirect.

TPS improves upon related network algorithms

Our comparison to TimeXNet, FunChisq, and the dynamic Bayesian network (DBN) demonstrates the limitations of these methods. TimeXNet divides all phosphorylated proteins into initial, intermediate, and late temporal bins based on the timing of their maximum phosphorylation change. Its network flow constraints require positive flow through the

intermediate nodes but permit the flow to travel from the intermediate nodes back to the initial nodes. This interpretation of the temporal data can lead to edge orientation errors.

The fundamental differences between TPS and related algorithms that detect statistical dependencies among phosphorylated proteins, such as correlation networks, FunChisq, and DBNs, is further emphasized by the simple artificial example. These competing approaches may capture many indirect effects and omit direct interactions that involve mechanisms besides phosphorylation. For example, in Figure 10A an edge is predicted between A and F although these proteins do not directly interact in the true pathway. These highly connected groups of nodes that contain false positive edges resemble the dense node clusters in the FunChisq network. In addition, B is excluded from the network because it participates via a different type of PTM and is not phosphorylated, and E is omitted despite being strongly differentially phosphorylated. When the valid protein temporal activity windows are restricted to only the time point of maximum phosphorylation change (Patil et al., 2013), the interaction from C to D is lost because C is no longer permitted to inhibit D at the first time point (Figure 10B). Furthermore, the interaction between E and G is oriented incorrectly.

TPS only asserts an edge direction when it is certain that all plausible pathway models must use an interaction in a particular direction. As in the NLP evaluation, this conservative approach leads to accurate direction predictions, which is crucial for designing validation experiments. When tested against the collection of EGFR reference pathways and all pathways in the BioCarta, Reactome, and PID databases, TPS makes only one incorrect direction prediction, though many of the directed edges are not in the reference pathways and cannot be confirmed

or refuted. In contrast, the DBN and TimeXNet predict more directed edges but make errors at a greater rate.

The reference pathway evaluation illuminates additional ways in which our constraint-based approach improves upon the existing methods. Unlike TPS, the DBN and FunChisq can infer only interactions between proteins observed in the phosphorylation time series data and ignore unphosphorylated nodes. Therefore, these methods miss important interactions involving proteins with no associated time series data, which TPS can detect. These include two of our experimentally-tested predictions (ABL2->CRK and AKT1->ZYX) and additional interactions with EGFR reference pathway members MAP2K1 and PAK1. Overall, the DBN predicts more edges than TPS, but fewer of those edges are in the reference pathways. FunChisq also produces a larger network than TPS but has no directed edges that overlap with directed edges in the reference pathways. Inspection of the FunChisq network reveals that this may be because its predicted network does not resemble the topology of the reference pathways. It contains several clusters of nodes, and nodes within a cluster are predicted to interact with most other nodes in the cluster.

By making greater use of the temporal information in the time series data, TPS can detect temporal inconsistencies in TimeXNet predictions that conflict with the reference pathway directions. We identified incorrectly oriented TimeXNet predictions, such as GRB2->EGFR and GRB2->ERBB2, for which TPS detects an inconsistency between the prediction and the input data. When providing the TimeXNet output as input to TPS, TPS reports that there is no pathway structure that can activate GRB2. Consequently, the GRB2 node and the incorrect

EGFR and ERBB2 interactions are removed. As an example of an invalid pathway structure, TimeXNet predicts that signal flows from LCK to CBL to GRB2. However, TPS shows that any directed path from LCK to GRB2 violates the phosphorylation timing because LCK is first differentially phosphorylated at 16 min, later than GRB2's initial change at 4 min. TPS detects similar contradictions for all other paths to GRB2 in the TimeXNet network. Although TimeXNet also models phosphorylation timing, it cannot model the dynamics of multiple phosphosites per protein and uses a coarse division of the time series data into three bins, which can produce misleading temporal interpretations.

Yeast osmotic stress response model

The mitogen-activated protein kinase Hog1 is a central regulator of the yeast hyperosmotic stress response, controlling osmo-responsive transcriptome changes (Chasman et al., 2014; O'Rourke and Herskowitz, 2004; Posas et al., 2000; Rep et al., 2000), transient translational inhibition (Melamed et al., 2008; Teige et al., 2001; Uesono and Toh-e, 2002), cell cycle arrest (Clotet et al., 2006; Duch et al., 2012; Escoté et al., 2004), and metabolic changes controlling intracellular osmolyte levels (Albertyn et al., 1994; Dihazi et al., 2004; Norbeck et al., 1996). The TPS pathway model correctly highlights Hog1's critical role in the phosphorylation response.

In our human EGF stimulation studies, our network models were constructed from a hybrid network that included general protein-protein interactions as well as kinase-substrate interactions. Therefore, the TPS pathway models may include adaptor and other classes of proteins that do not directly induce substrate phosphorylation changes but are nevertheless relevant to the overall pathway. In contrast, in yeast there are sufficient known kinase-substrate

and phosphatase-substrate interactions (Chasman et al., 2014; Sharifpoor et al., 2011), so we apply TPS to a network consisting primarily of directed post-translational modification interactions and forgo general protein-protein interactions. We expected that the resulting TPS model will be more directly relevant to the extremely rapid phosphorylation changes and suggest specific mechanistic hypotheses at the expense of excluding come classes of HOG pathway members and failing to include some of the phosphorylated proteins.

Limitations of the phosphoproteomic data as well as false positives and false negatives in the background network explain some of the divergence between the TPS pathway model and the canonical HOG pathway model. Several membrane proteins and other upstream members in the HOG pathway that are excluded from the TPS model – Cdc42, Hkr1, Msb2, Opy2, Ste11, and Ypd1 – were not detected or not differentially phosphorylated in the phosphoproteomic data. This may be because it only profiled the first minute of the osmotic stress response. The interaction Hog1->Msn4 was absent from the background network, leading TPS to search for an alternative connection to Msn4. The background network also included interactions that are likely false positives such as Ste50->Pbs2, Ste50->Ssk2, and Rck2->Pbs2, which were selected by TPS. The Rck2->Pbs2 edge is contained in the yeast Kinase Interaction Database (Sharifpoor et al., 2011) but is not reported in the study from which it was originally curated (Bilsland-Marchesan et al., 2000). The database curators corrected this false positive edge after we notified them.

Most of the predicted Rck2 targets in the TPS network (54%) were supported by the *RCK2* mutant study (Romanov et al., 2017). Only seven of the 193 (4%) proteins that were quantified

in the Kanshin et al. osmotic stress study and have decreased phosphorylation in the *RCK2* mutant strain were predicted as direct Rck2 targets. This is because both direct and indirect Rck2 targets are affected by its deletion.

Our network made many predictions on how yeast coordinate cell division during acclimation to osmotic stress. In response to severe osmotic stress, Hog1 induces temporary arrest in G1, G2, or S-phases (Clotet et al., 2006; Escoté et al., 2004; Yaakov et al., 2009), allowing cells to repair damage before dividing. Cdc28 was identified as a major hub in the TPS network, indicating Cdc28 regulation may play a crucial role in this response. To test these predictions, we compared the TPS model to two phosphoproteomic studies performed on chemically inhibited *cdc28-as* strains under nutrient rich and cell cycle arrest conditions (Holt et al., 2009; Kanshin et al., 2017). Sixty-seven out of ninety-one (74%) predicted Cdc28 targets have at least one phosphosite with defective phosphorylation in *cdc28-as*. These 67 proteins represent 18% of the 364 proteins that were observed in the osmotic stress response study and have a defective phospohosite in *cdc28-as*. Despite the strong support for most of the predicted Cdc28 targets, confirming Cdc28's role in the early osmotic stress response requires further study. Both detected Cdc28 sites did not significantly increase or decrease in phosphorylation (Kanshin et al., 2015). Prior studies demonstrated Cdc28 is inhibited after osmotic shock (Clotet et al., 2006) but at later time points, leaving the possibility that Cdc28 activates its direct targets in the first minute of the response before it is inhibited.

TPS scales to large data sets

We assessed the scalability of TPS by running it on randomly generated time course data and graphs and found that TPS can scale to graphs with more than 100,000 edges and 100,000 phosphosites. TPS performs similarly on our EGF case study data and randomly generated data of comparable size (graphs with 750 edges) in terms of runtime. We computed the median TPS running time (in seconds) in terms of the number of input graph edges in log-log scale for three run replicates per input size. We observe that TPS takes approximately 13 hours to run on an input graph with 128,000 edges and 108,711 phosphosites.

TPS predictions are robust against varying the p-value threshold for detecting significant changes

We evaluated the effects of changing the p-value threshold for detecting significant changes in the time course data by analyzing inferences made using different threshold values (Table 9). As we decrease the p-value threshold from the default of 0.01, fewer time points are considered as significant changes for a peptide. With fewer significant time points, TPS makes fewer signed, directed predictions. On the other hand, increasing the p-value threshold from its default value leads to an increased number time points with significant changes. This in turn leads to a greater number of signed, directed predictions, until it introduces too much ambiguity to the time series interpretation to make unambiguous signed, directed predictions. We observe that TPS makes a varying number of signed, directed predictions depending on the p-value threshold, but the predictions made are consistent across the range of threshold values.

Meanwhile, the unambiguously directed predictions are mostly stable across different threshold values.

Author contributions

Conceptualization, A.S.K., S.S., R.B., A.W.-Y., E.F., J.F., and A.G.; Methodology, A.S.K., K.B., R.B., A.W.-Y., E.F., J.F., and A.G.; Software, A.S.K. and A.G.; Validation, K.B. and N.D.C.; Formal Analysis, A.S.K., D.R.C., and A.G.; Investigation, A.S.K., K.B., D.R.C., A.M., N.D.C., M.E.M., and A.G.; Data Curation, A.S.K., K.B., D.R.C., A.W.-Y., and A.G.; Writing, A.S.K., K.B., M.E.M., R.B., A.W.-Y., E.F., J.F., and A.G.; Visualization, A.S.K., E.F., and A.G.; Supervision, R.B., A.W.-Y., E.F., J.F., and A.G.

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